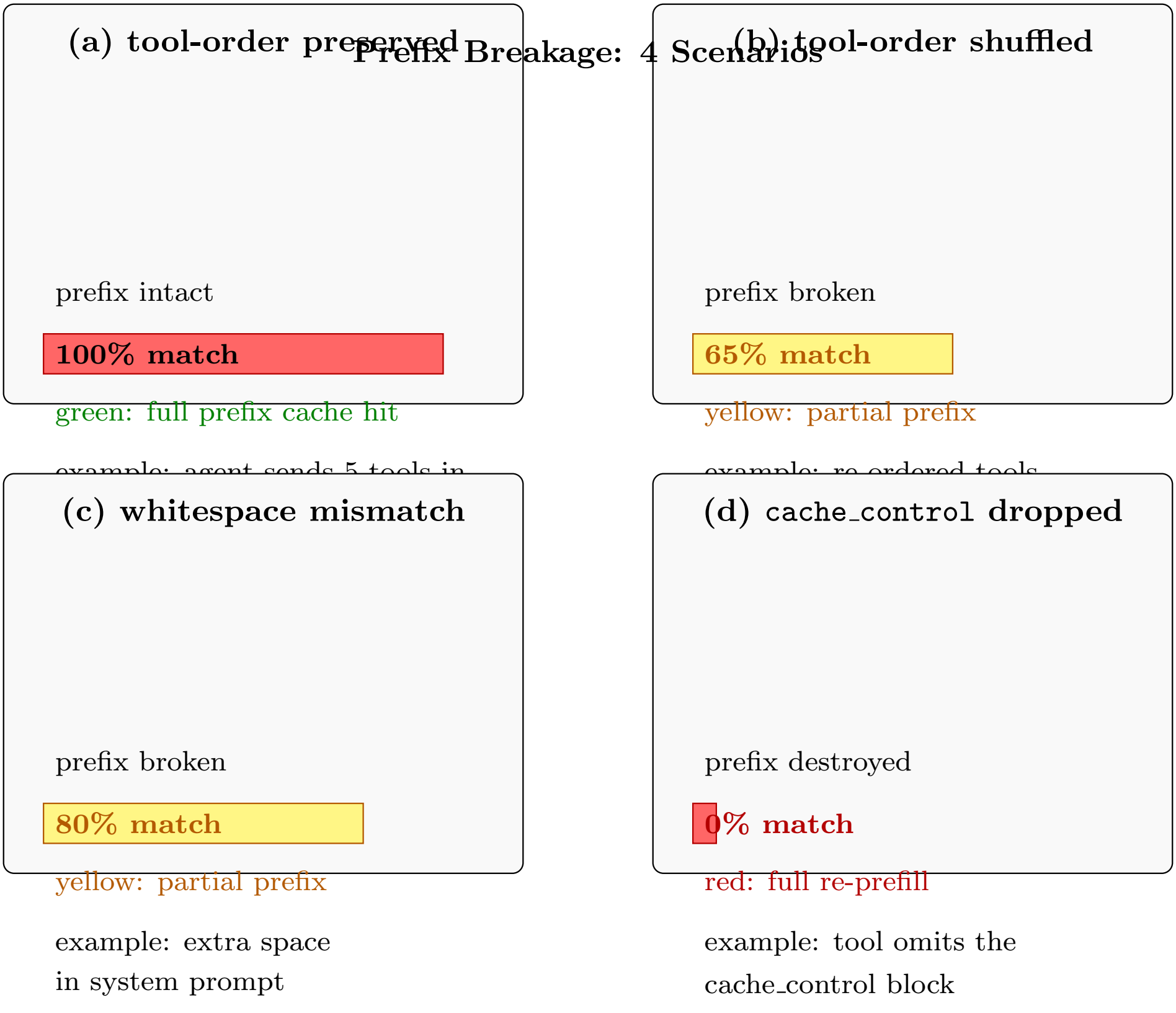
Prefix Breakage: 4 Scenarios
(a) tool-order preserved
prefix intact
100% match
green: full prefix cache hit
example: agent sends 5 tools in
(b) tool-order shuffled
prefix broken
65% match
yellow: partial prefix
example: re-ordered tools
(c) whitespace mismatch
prefix broken
80% match
yellow: partial prefix
example: extra space
in system prompt
(d) cache_control dropped
prefix destroyed
0% match
red: full re-prefill
example: tool omits the
cache_control block